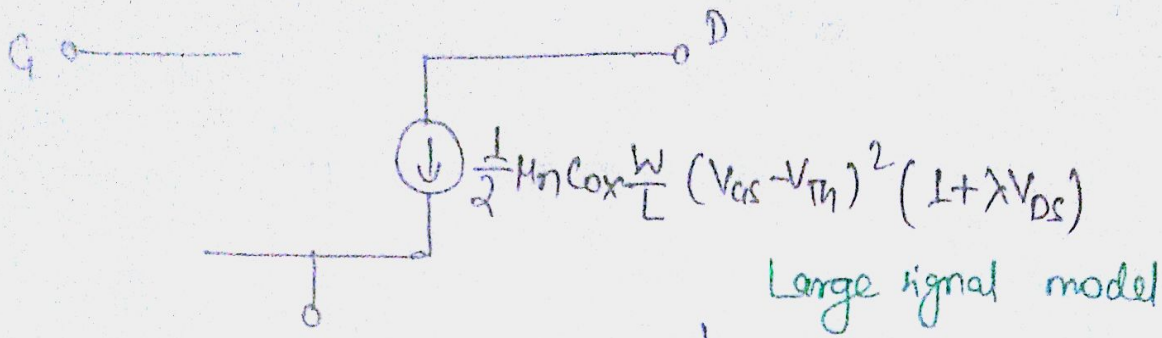


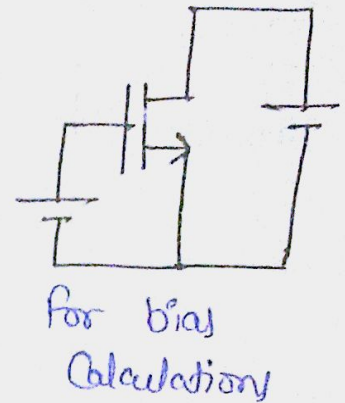
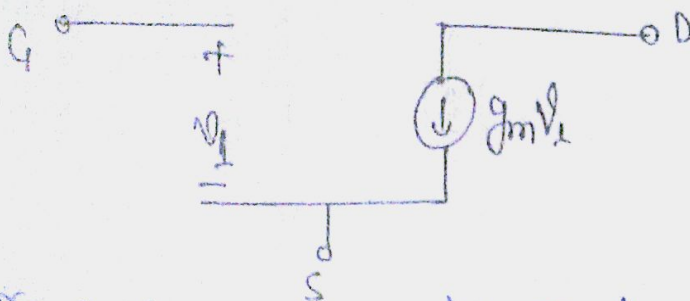
Lecture - 34

Large and small-signal models:-

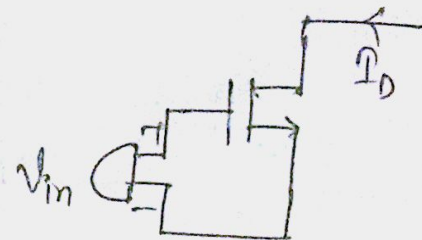
(In saturation)



For small signal model,

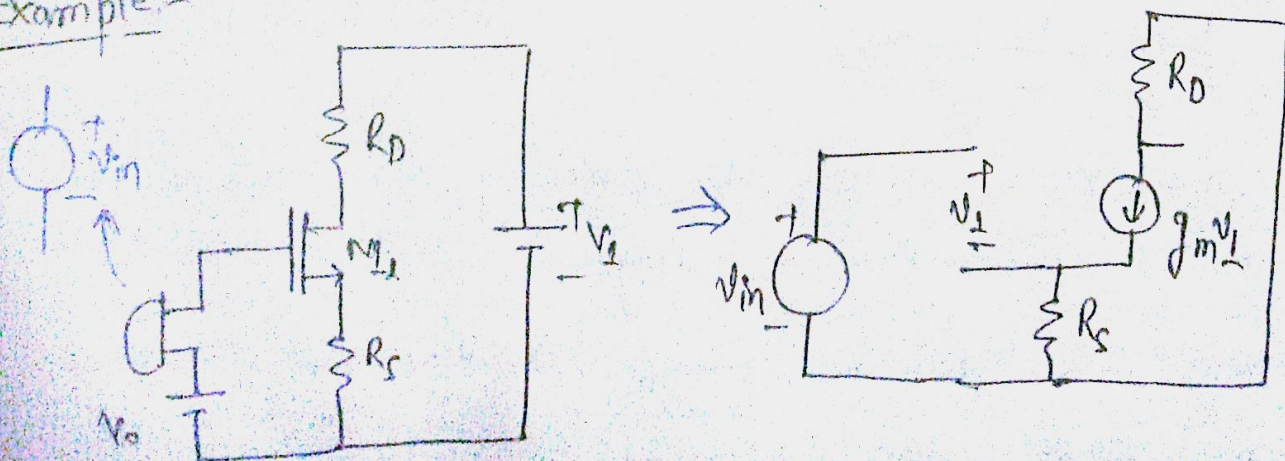


If in the circuit, the signal comes in which is small then



* Small signal model of constant sources (not changing with time) is replaced by zero i.e. voltage source is replaced by short ckt and current sources are replaced by open ckt.

Example:-

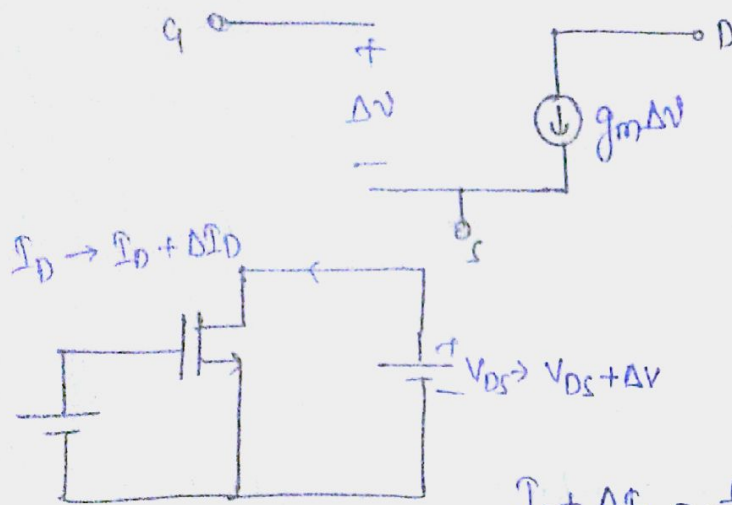


* General procedure of constructing a small signal model:-

- ① Apply proper bias voltages to the device.



- ② Increment the voltage difference between two terminals.
- ③ Measure all currents increments.
- ④ Model this change by a proper electrical device.



* when we change voltage at gate & source, we did not feel channel length modulation because it is caused by change in V_{DS} .

$$I_D + \Delta I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{th})^2 (1 + \lambda V_{DS} + \lambda \Delta V)$$

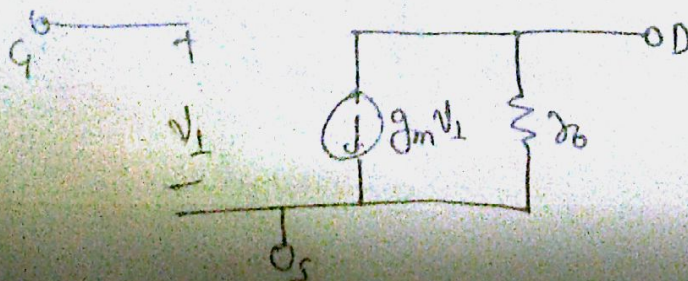
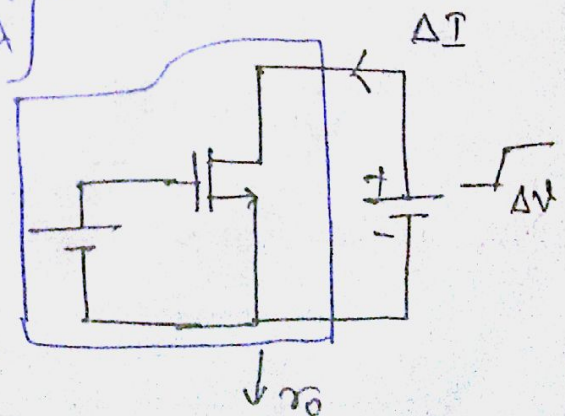
$$I_D + \Delta I_D = I_D + \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{th})^2 (\lambda \Delta V)$$

$$\Rightarrow \Delta I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{th})^2 (\lambda \Delta V)$$

$$\Rightarrow \left(\frac{\Delta V}{\Delta I} \right) = \left[\frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{th})^2 \lambda \right]^{-1}$$

$\approx I_D$

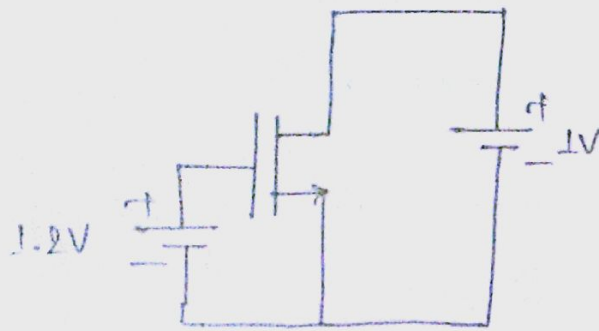
$$\Rightarrow \frac{\Delta V}{\Delta I} \approx \frac{1}{\lambda I_D} = r_o$$



* When we want to do bias calculations we typically neglect the channel length modulation and then when we model small signal, our first reaction is to neglect channel length modulation.

* We neglect the effect of channel length modulation on the g_m expression.

Example:-



$$\mu_n C_{ox} = 100 \mu A/V^2$$

$$V_{th} = 0.5 V$$

$$\lambda = 0.2 V^{-1}$$

$$\frac{W}{L} = 20$$

$$V_{DS} = 1V, \quad V_{GS} - V_{th} = 1.2 - 0.5 = 0.7V$$

$$\Rightarrow V_{DS} > V_{GS} - V_{th}$$

Therefore, it is in saturation.

$$g_m = \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{th}) = 0.7 mS = \frac{1}{1.43 k\Omega}$$

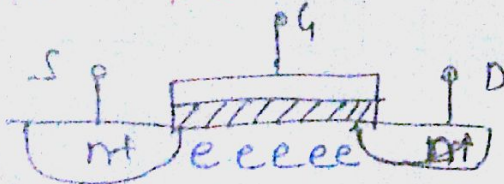
$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{th})^2 (1 + \lambda V_{DS})$$

$$= 0.245 mA$$

$$r_o = \frac{1}{\lambda I_D} = \frac{1}{0.2 \times 0.245} = 20.4 k\Omega$$

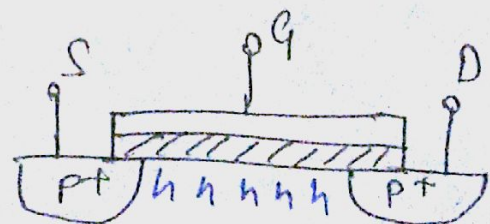
$$g_m r_o \approx 14$$

MOS Device Types:-



p-substrate

NMOS



n-substrate

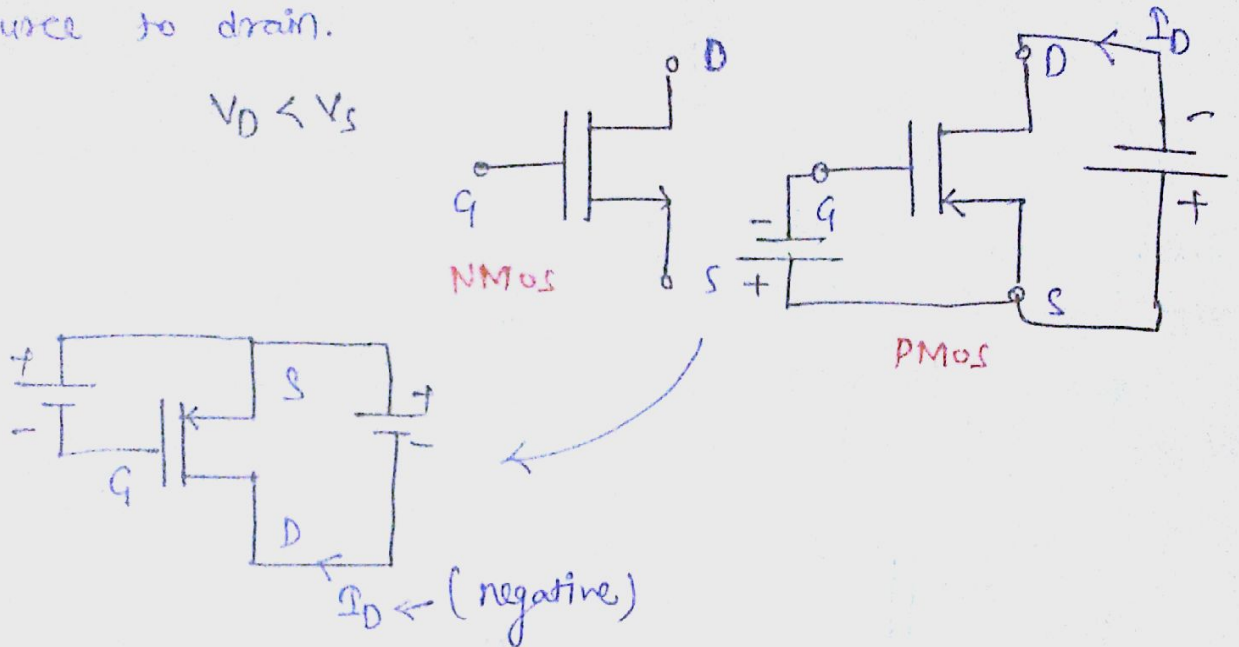
PMOS

* In PMOS, holes create a channel. Therefore Gate has to be negative with respect to the source by one threshold so that channel can be formed.

$$V_G < V_S, \quad V_{Th} < 0$$

* The source is more positive than drain therefore current flow from source to drain.

$$V_D < V_S$$

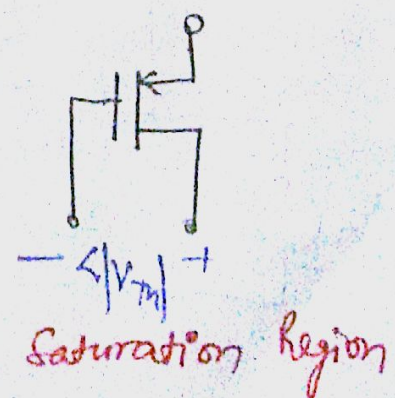
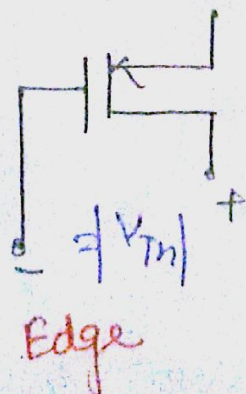
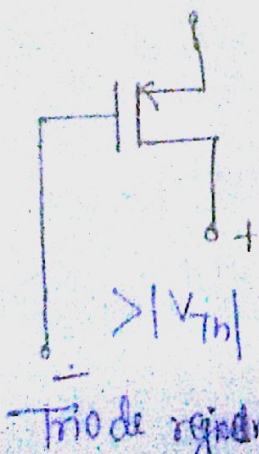
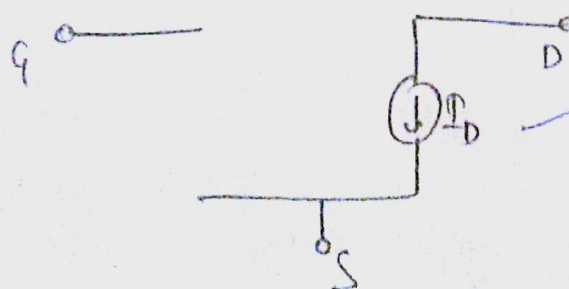


In saturation,

$$I_D = -\frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{Th})^2 (1 - \lambda V_{DS})$$

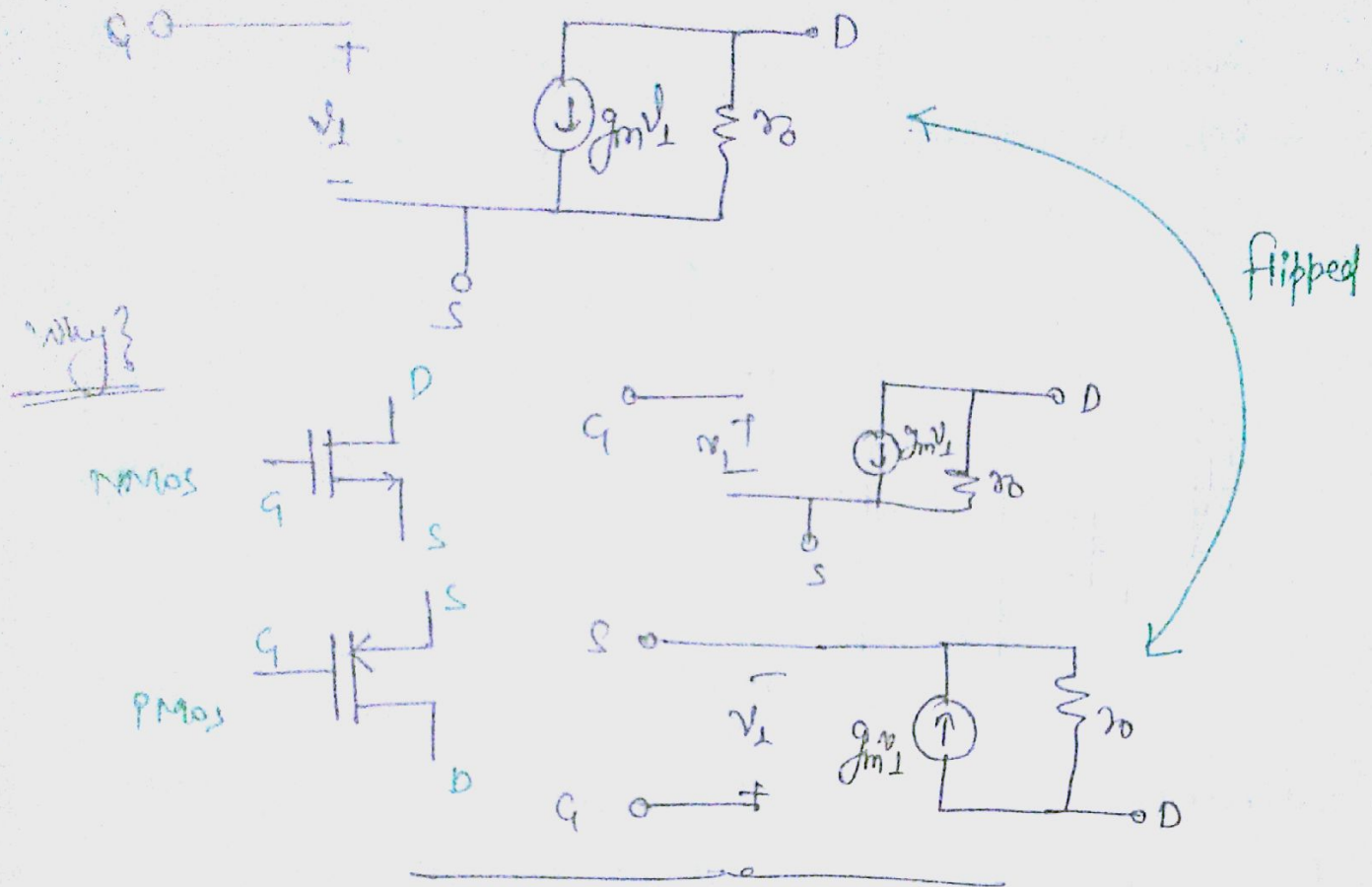
$\lambda > 0$

Large signal model of PMOS device:-

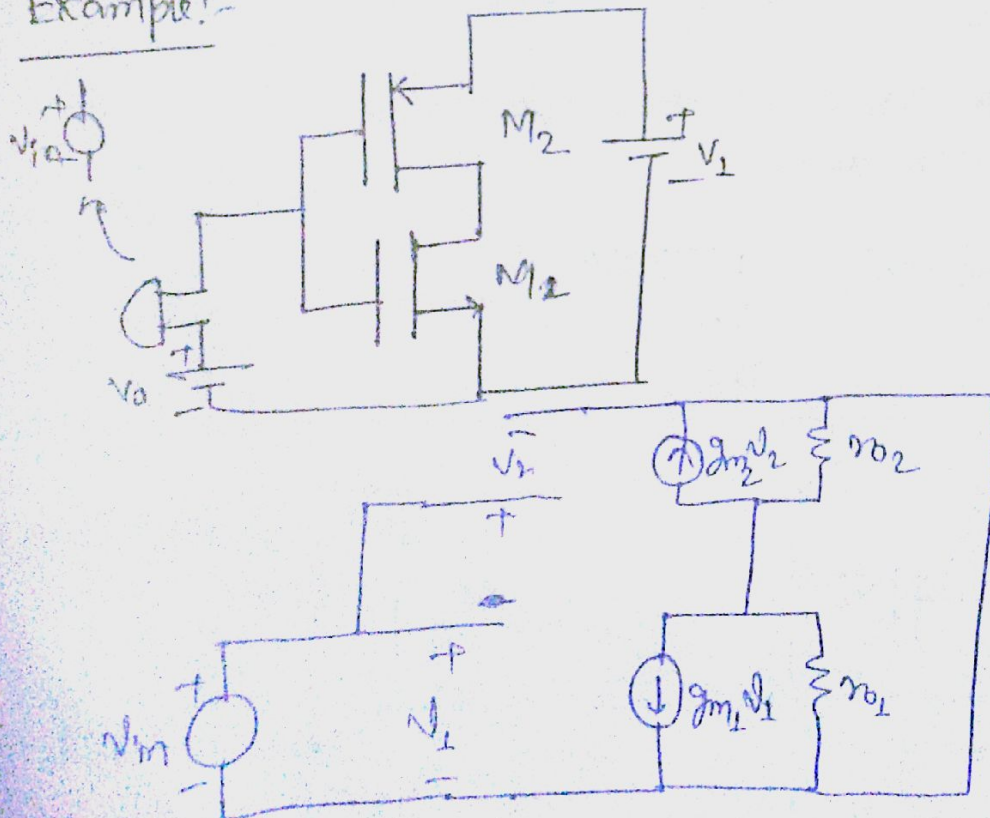


Small signal model of Pmos device:-

Identical to the small signal model of NMOS device.

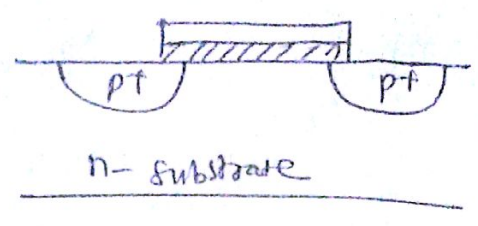
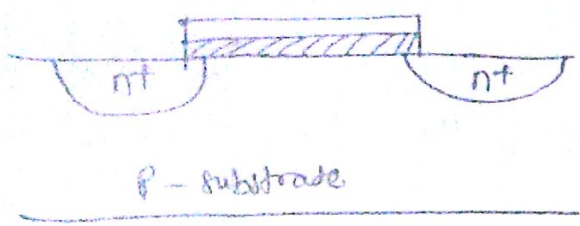


Example:-

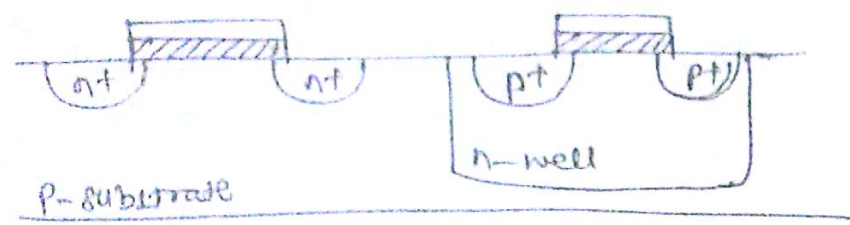


if v_1 and v_2 are equal.
 $\uparrow$ $g_{m1}v_1$ and $g_{m2}v_2$ are
 in parallel.
 $\uparrow$ r_{o1} and r_{o2} are
 in parallel.

CMOS Technology:- CMOS (Complementary MOS) means that we would like to have both NMOS & PMOS devices in our circuit.

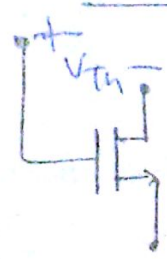


* For making CMOS, we either have n type wafer or p-type wafer.



CMOS using N-well

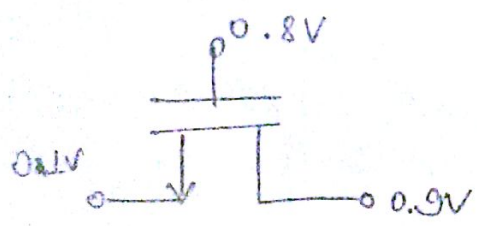
Example:-



At the edge

$$V_{thN} = 0.5V$$

$$V_{thP} = -0.6V$$



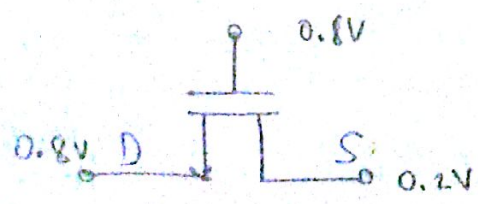
$$V_{GS} = 0.7V$$

$$V_{GS} - V_{th} = 0.2V$$

$$V_{DS} = 0.8V$$

$$\Rightarrow V_{DS} > V_{GS} - V_{th}$$

$\Rightarrow$ Saturation (device is ON)



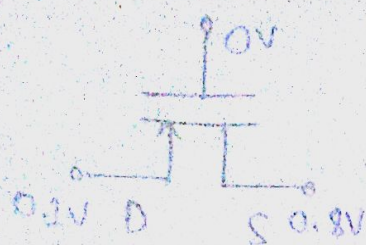
$$V_{GS} = 0.6V$$

$$V_{GS} - V_{th} = 0.1$$

$$V_{DS} = 0.9$$

$$V_{DS} > V_{GS} - V_{th} \Rightarrow \text{Saturation}$$

~~At the edge~~ The terminal which has lower potential is source terminal it's not depend on arrow. (NMOS)



* In PMOS, source must be more positive than drain.

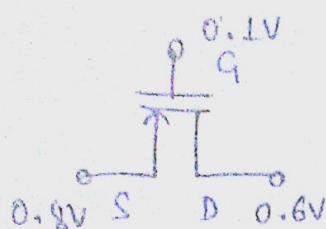
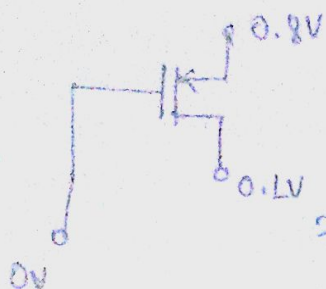
$$V_{GS} = -0.8V$$

$$V_{GS} - V_{Thp} = -0.8 + 0.6 = -0.2V$$

$$V_{DS} = -0.7$$

$$\Rightarrow V_{DS} < V_{GS} - V_{Th} \quad \leftarrow \text{PMOS}$$

saturation



$$V_{GS} = -0.7V$$

$$V_{DS} = -0.2V$$

$$V_{GS} - V_{Thp} = -0.7 + 0.6 = -0.1$$

$$V_{GS} - V_{Th} > V_{DS} \Rightarrow \text{saturation}$$

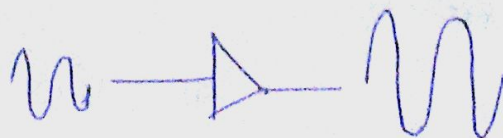
* CMOS Amplifiers:-

* Amplifier design procedure:-

- ① Select an amplifier topology.
- ② Bias the transistors to obtain proper values for g_m , r_o , ...
- ③ Determine the characteristics of the circuit of the amplifier.

* Amplifier characteristics:-

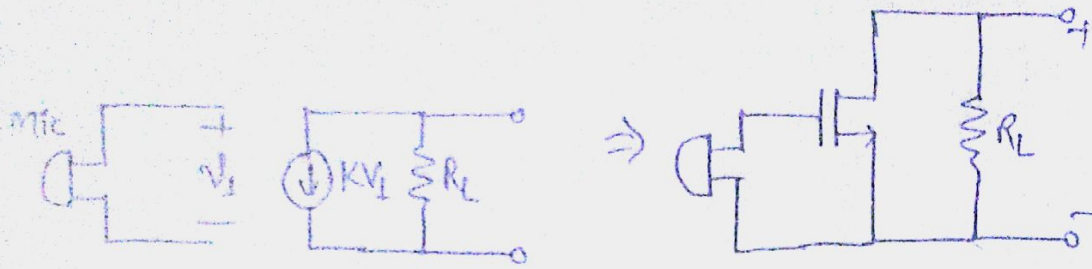
- ① Gain
voltage gain A_v



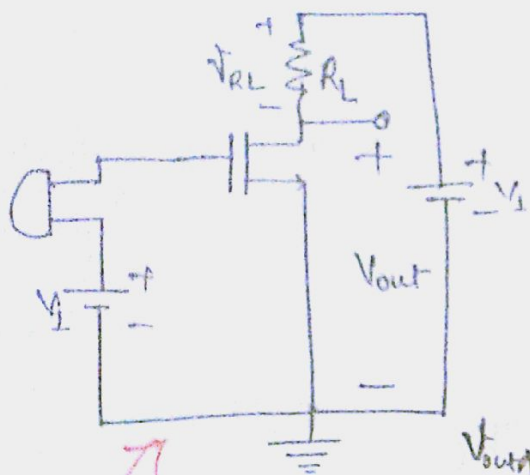
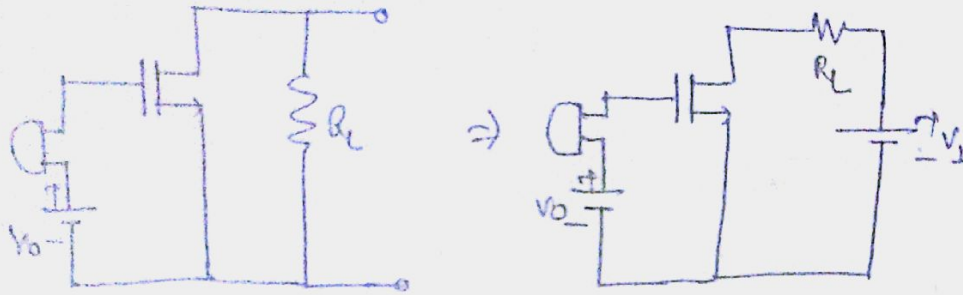
→ Gain is the ratio of output voltage swing to input voltage swing.

- ② Power dissipation
and many more

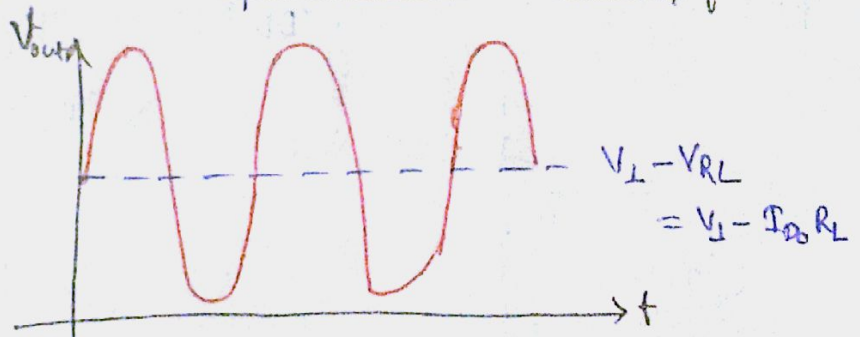
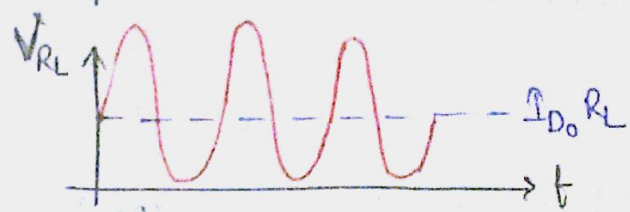
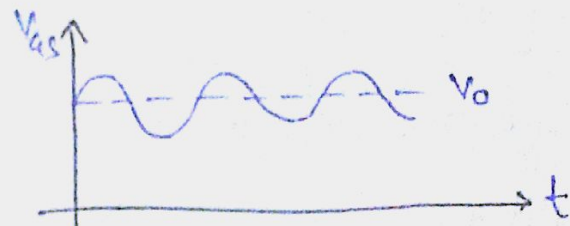
* Let's build an amplifier: -



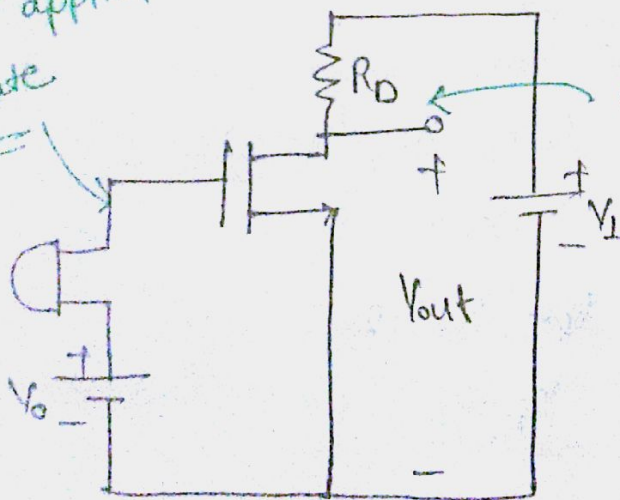
Evolution of our amplifier



Common Source



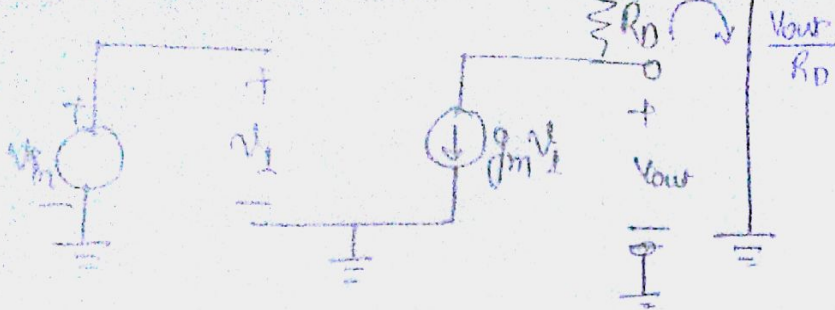
Input applied to gate



output sensed at drain

Common Source Amplifier

* In common source Amplifier, input is applied to gate and output is sensed at drain.



$$v_{in} = v_1$$

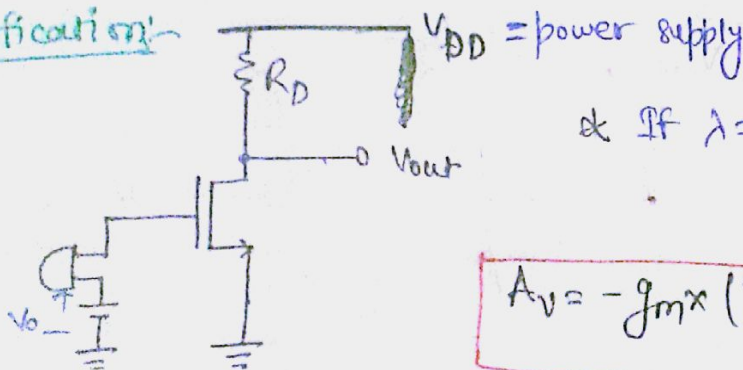
$$\therefore \frac{v_{out}}{R_D} = -g_m v_1 \Rightarrow \frac{v_{out}}{R_D} = -g_m v_{in}$$

$$\boxed{\frac{v_{out}}{v_{in}} = -g_m R_D} \leftarrow \text{Voltage gain of simple common source topology.}$$

This -ve sign indicates that output voltage is flipped of input in larger amplitude.

Lecture - 36

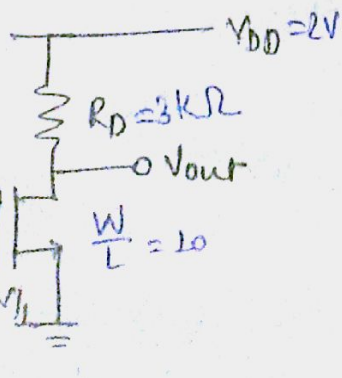
Simplification:-



$$\text{If } \lambda = 0, A_v = -g_m R_D$$

$$\boxed{A_v = -g_m \times (\text{Total resistance tied between drain \& ac ground})}$$

Example:-



$$\mu_n C_{ox} = 100 \mu A/V^2, V_{th} = 0.5V$$

$$\lambda = 0$$

$$V_{gs} = 1.2V$$

$$V_{gs} - V_{th} = 0.7V$$

$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{gs} - V_{th})^2 = 0.245 \text{ mA}$$

① Bias Calculations:

Assume sat. region

$$V_{DD} - V_{DS} = I_D R_D$$

$$\Rightarrow V_{DD} - I_D R_D = V_{DS} \Rightarrow V_{DS} = 2 - 0.735 = 1.265V$$

$V_{GS} > V_{GS} - V_{TH}$ so if it's in saturation

② Small signal parameters:-

$$g_m = \frac{2I_D}{V_{GS} - V_{TH}} = \frac{0.49}{0.7} = 0.7 \text{ mS} = \frac{1}{1.43 \text{ k}\Omega}$$

③ Calculate the gain:

$$A_v = -g_m \times (R_D) = -\frac{1}{1.43} \times 3 = -2.1$$

Need to double the gain, Try doubling R_D :-

→ If we double R_D then voltage drop across it doubles as current I_D is not changed,

$$V_{DS} = V_{DS} - I_D R_D = 0.53 \text{ V}$$

$$V_{GS} - V_{TH} > V_{DS}$$

So it can occur in triode or linear region.

Therefore it does not work.

• Try increasing g_m :-

$$g_m = \sqrt{2\mu_n C_{ox} \frac{W}{L} I_D} \rightarrow \begin{matrix} I_D \uparrow \\ \frac{W}{L} \uparrow \end{matrix}$$

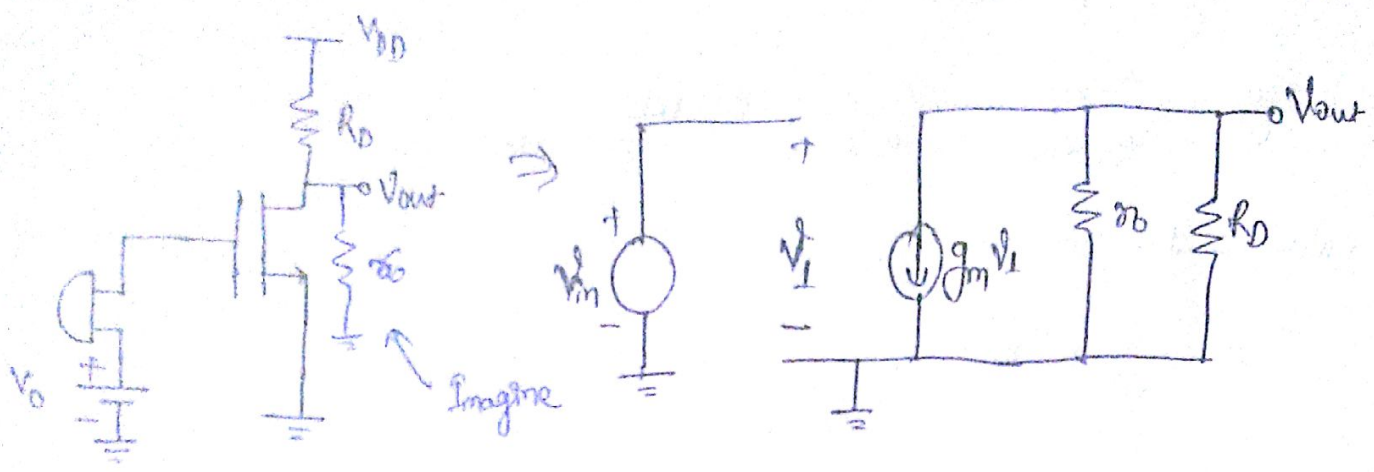
→ If I_D goes up then previous result will occur so it does not work. Because eventually triode.

→ Now if we take I_D constant & increase $\frac{W}{L}$ then we do this by decreasing $V_{GS} - V_{TH}$ and this equation does not hold.

$$g_m \approx \frac{I_D}{1.5V_T} \rightarrow 26 \text{ mV}$$

So it does not work because gain is bounded it does not go beyond the limit.

Inclusion of channel length modulation:-

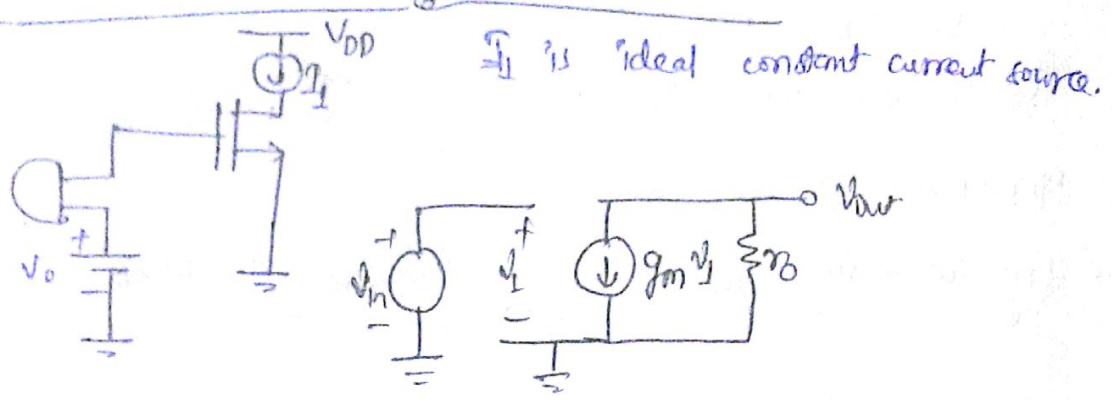


$v_i = v_m$

$$A_v = -g_m \times (r_o || R_D)$$

Total resistance tied b/w drain and ac ground

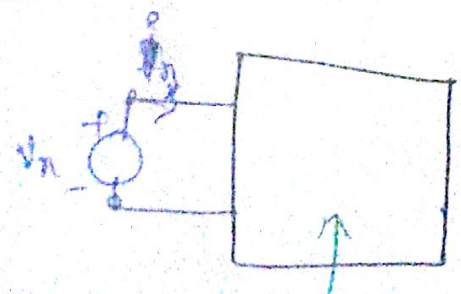
Example:-



$A_v = -g_m r_o \approx -5 \text{ to } -80$
Intrinsic gain of MOS device

→ The intrinsic gain shows that best voltage gain we can get from a transistor, we cannot get higher than that.

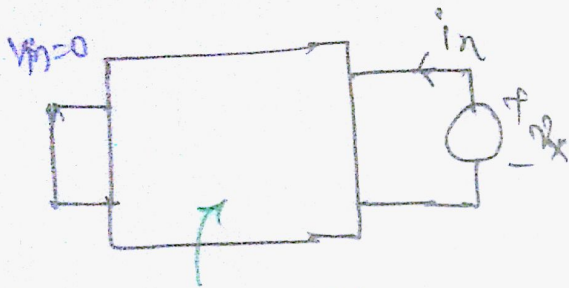
* Concept of Port impedance:-



* We kill all the independent source i.e. set to zero.

Input impedance, $R_{in} = \frac{v_{in}}{i_{in}}$

Set all independent sources to zero.



Output
impedance

$$R_{out} = \frac{V_x}{I_x}$$

set all independent
sources to zero

* It is same as evaluating Thevenin's resistance.